When is committee force uncertainty worth computing? Decision quality, cheap gates, and the baselines that must be beaten

Ian Poon

Harvard University

Abstract

Ensembles of machine-learned interatomic potentials are the standard tool for deciding which structures to label next, and the standard way to validate a cheap approximation of one is to check that it reproduces the exact ensemble. That is the wrong target. We separate two questions that the literature conflates: whether exact committee disagreement predicts reference-force error at all (*oracle quality*), and what a cheap screening gate costs relative to whichever oracle one has chosen (*preservation*). Preservation is independent of oracle quality, which makes the second claim robust to the first failing. On the first question, exact committee disagreement ranks top-5% force error at AUROC 0.737–0.848 across six molecular systems and 0.798 across 136,923 periodic crystals evaluated with a frozen foundation-model trunk, against a random null of 0.499. A sum-over-coordinates (*global*) statistic beats the max-component rule used in prior work in 24 of 24 cells—but only at fixed system size, because it is extensive; at variable size it acquires structures 2.50× larger than the intensive alternative and the ordering reverses. On the second, a *deterministic* leading-subspace estimator skips 0.822–0.943 of exact evaluations while preserving the exact policy's decisions, and beats a randomised low-rank sketch at matched cost on 5 of 6 systems; an exact finite-sample law for the randomised estimator predicts its measured failure to a mean absolute error of 0.015 with no free parameters. The result practitioners should act on is the least flattering. On a real materials pool the max mean-force norm—free, already computed in the forward pass—reaches AUROC 0.943 against the committee's 0.798, and adding the committee on top of it changes that by 0.0002 — not significant. On fixed-composition systems the committee does add significantly, on 8 of 10 systems by 0.022–0.243 AUROC. Committee uncertainty is therefore not redundant in general; it is redundant where a free signal is already near-saturated, and every future claim should be scored against that signal rather than against chance.

1 Introduction

Machine-learned interatomic potentials (MLIPs) are trained on labels that cost orders of magnitude more than the model does to evaluate, so which structures get labelled matters more than almost any modelling choice. The dominant selection heuristic is query-by-committee [10]: train an ensemble, evaluate the spread of its predictions, and label where the spread is largest. For force-field models the natural spread is over *forces*, and obtaining it exactly costs one reverse-mode pass per ensemble member.

A multi-head committee makes that cheaper in principle [3]. Instead of M independent models, one trunk carries M output heads, so a single reverse pass through the trunk with an appropriate head-space cotangent yields a projection of the per-head force matrix. Whether a small number of such projections suffices to reproduce the acquisition decisions of the exact committee is the question that a sketching analysis is usually asked to answer.

We argue that this is the second question, and that asking it first has hidden a weakness in the whole approach. The literature validates a cheap estimator against the *exact committee*, treating the committee as ground truth. But the committee is itself a proxy—for the model's error against the reference method—and it is that error a practitioner wants to find. So we decompose:

- **Factor A, oracle quality.** Does exact committee force disagreement rank reference-force error? This is answerable offline from cached per-head forces and requires no sketching at all.
- **Factor B, preservation.** Given whichever oracle one has chosen, how many of its decisions does a cheap gate reproduce, and at what cost?

Factor B is logically independent of Factor A: *whatever you decided to trust, we reproduce it more cheaply* is a claim that survives Factor A being mediocre. Reporting them together, rather than reporting only the sketch fidelity, is the paper's main methodological contribution, and it is what surfaces the result in Section 7—that on a realistic materials pool the committee loses to a signal that costs nothing.

Every numeric claim below is derived programmatically from a committed record; none is transcribed. Section 9 describes the reproduction path.

2 Method

2.1 Centred head space

Let $F(x) \in \mathbb{R}^{3N\times M}$ hold the forces predicted by M committee heads for a structure x with N atoms. Acquisition scores depend only on the spread across heads, so the mean-force direction is separated out: with Q_c an $M \times r$ orthonormal (Helmert) basis of the centred subspace, $r = M-1$, the quantity of interest is $A(x) = Q_c^\top F(x)^\top F(x) Q_c$, an $r \times r$ matrix independent of N. The per-coordinate variance is $v_d = \|Q_c^\top F_d\|^2/(M-1)$.

2.2 Three acquisition statistics

Prior work ranks structures by the largest per-coordinate standard deviation, $\max_d \sqrt{v_d}$ (*max-component*). We also consider the largest per-atom value (*max-atom*) and the sum $\sum_d v_d$ (*global*). The global statistic is a trace of $A(x)$ and so is by far the easiest to estimate from few projections; whether it is also the better physical predictor was pre-registered as a secondary hypothesis before any of it was run.

2.3 Estimators

A projection (*lane*) costs one reverse pass and returns $g_k = F w_k$ for a head-space direction w_k. We compare Haar and Gaussian sketches, head subsampling, a low-rank control variate that evaluates r_0 leading directions exactly and sketches the residual—the split that Meyer et al. [9] show is optimal for trace estimation—and a purely deterministic *leading-only* estimator that spends every lane on exact leading eigendirections of the design-set Gram and discards the residual entirely.

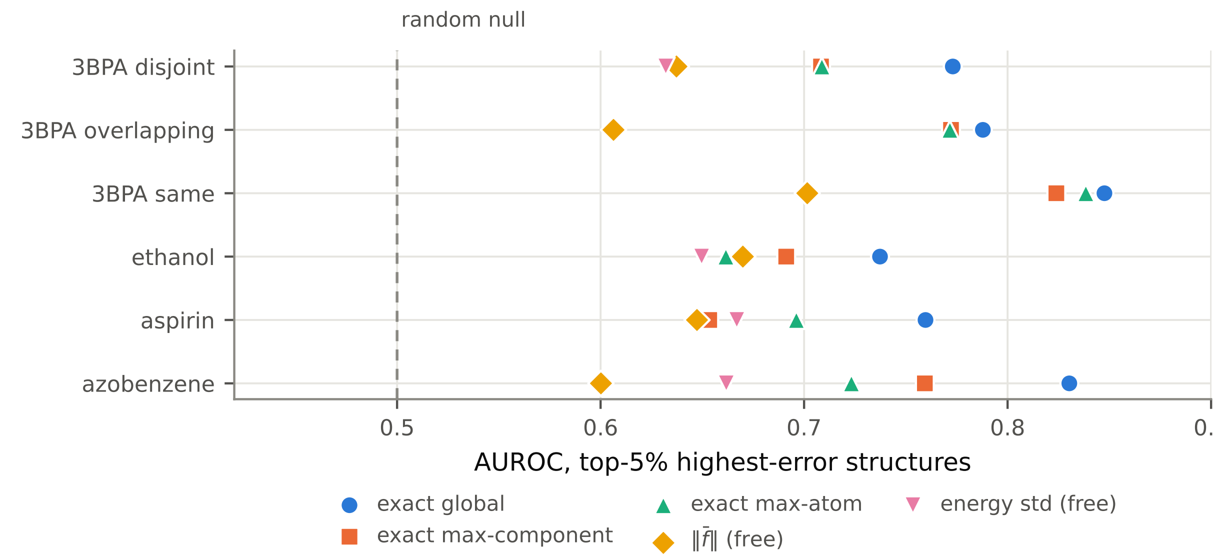

Figure 1: Exact committee disagreement ranks reference-force error, and the global statistic ranks it best at fixed system size. AUROC for detecting the top-5% highest-error structures, by signal, on the six molecular systems. The dashed line is the random null; every signal carries a distinct marker so the comparison survives grayscale printing. The two *free* signals cost no backward pass, and the margin over the better of them is 0.067–0.182 AUROC.

2.4 The calibrated gate

The gate skips the exact computation when a cheap score $\hat{S}$ times a calibration constant falls below a threshold. The constant is the split-conformal one: c_α is the $\lceil (n+1)(1-\alpha) \rceil$-th order statistic of the calibration ratios $S/\hat{S}$ [8, 11]. When $\lceil (n+1)(1-\alpha) \rceil > n$ no order statistic carries the finite-sample guarantee and the implementation returns $+\infty$, which makes the gate skip nothing—the safe failure. Realised coverage is verified against the $\mathrm{Beta}(k, n{+}1{-}k)$ law by simulation rather than asserted.

Design, calibration and test roles are disjoint contiguous blocks with guard bands sized from a measured integrated autocorrelation time; every split is content-hashed and recomputes from its recorded seed. Confidence intervals use a paired moving-block bootstrap [7].

3 Does committee disagreement predict reference-force error?

Across six molecular systems—3BPA [6] at 1200 K under three committee constructions, and three rMD17 molecules [4]—exact global disagreement ranks the top-5% highest-error structures at AUROC 0.737–0.848, with a random null at chance. The oracle is *moderate*: useful, and far from perfect. Nothing is fitted in this analysis, so there is no leakage risk.

3.1 The global statistic wins—at fixed system size

The global statistic beats max-component in 24 of 24 (system $\times$ error-score) cells, 19 of them significantly at 95%, and in no cell is it significantly worse (Figure 1). Because the global statistic is also the one a sketch estimates most easily, the negative result about small sketches (Section 5) applies to the statistic one should not have been using.

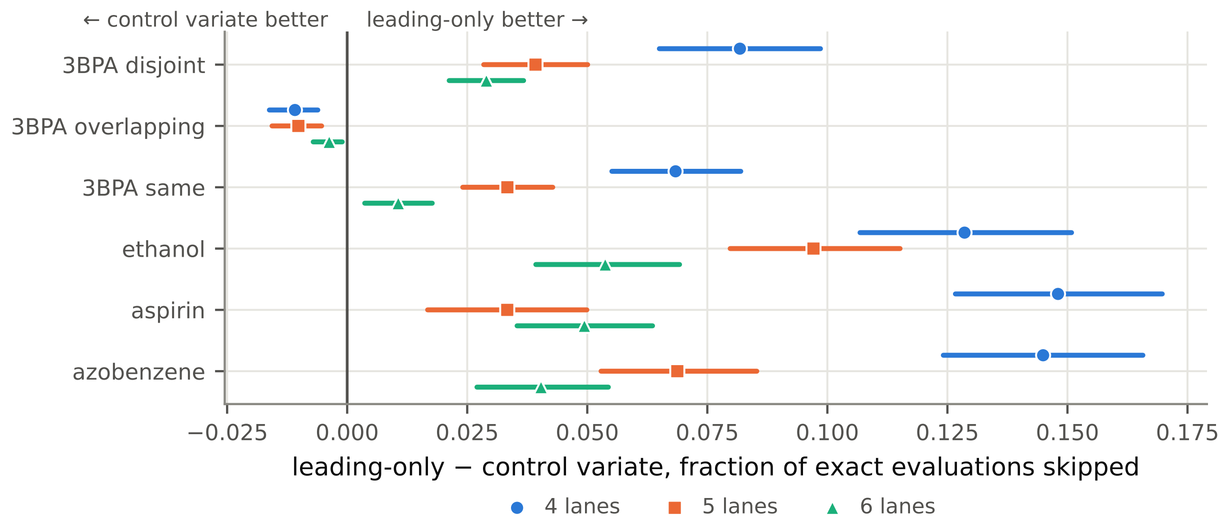

Figure 2: Randomisation does not earn its lanes. Difference in the fraction of exact evaluations skipped between the deterministic leading-only estimator and the randomised control variate, at three matched lane budgets, with 95% paired block-bootstrap intervals. Positive is leading-only better. It wins significantly on 5 of 6 systems at every budget; the single exception is 3BPA overlapping, in a regime where both estimators already skip more than 93%.

This holds *at fixed* N, and the qualifier is load-bearing rather than cosmetic; Section 7.2 shows the ordering reverse when system size varies, and why it could not have shown up here.

3.2 The margin over free signals

Head-energy spread costs one forward pass; the norm of the mean force costs nothing at all. Both are informative (AUROC 0.60–0.70). Exact global disagreement beats the better of them by 0.067–0.182 AUROC, and pairwise the advantage is significant in 9 of 10 comparisons. So the backward passes buy something real but not overwhelming—and Section 7 shows a regime where they buy nothing.

4 What the cheap gate costs

Evaluated on held-out test blocks, the gate skips 0.822–0.943 of exact evaluations while reproducing the exact policy's decisions. We report this count rather than a speedup because it is independent of hardware, autodiff backend and batching.

4.1 Randomisation does not earn its lanes

At matched lane budget the deterministic leading-only estimator beats the randomised control variate on 5 of 6 systems, at every budget tested, and has zero probe-seed variance by construction (Figure 2). The mechanism is specific to calibrated gates: c_α is fitted to the ratio $S/\hat{S}$, so a systematic bias in $\hat{S}$ is absorbed exactly, and only its *variance* costs skip fraction. Leading-only is biased low by construction and pays nothing for it. Since the head-space spectrum is concentrated—stable rank $\approx$ 1.2–4.3 of a possible 7—a random projection is being asked to do the one thing concentration makes hard.

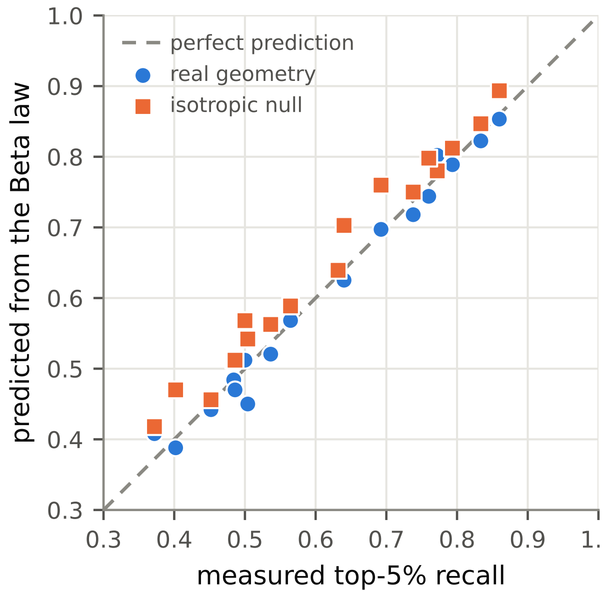

Figure 3: The finite-sample law predicts the negative result with no free parameters. Predicted against measured top-5% recall for six systems at three sketch budgets. Using the real head-space geometry gives a mean absolute error of 0.015; the isotropic null is both worse and systematically optimistic, which is why coordinate coupling cannot be neglected.

The practical consequence is that the method gets simpler: at fixed budget, spend every lane on exact leading directions.

5 Theory: the negative result is predictable

For a Haar sketch of K of the r centred directions, the ratio of estimated to true per-coordinate variance is distributed exactly as $\hat{v}_d/v_d \sim (r/K)\,\mathrm{Beta}(K/2,(r-K)/2)$. Injecting precisely this noise into the exact variances and recomputing top-5% recall reproduces the measured recall across six systems and three budgets to a mean absolute error of 0.015, with no free parameters (Figure 3).

Two corrections to the mechanism previously offered follow. First, a common multiplicative bias cannot change a rank statistic, so the damage is estimator *variance*, not bias. Second, the coordinate directions cannot be treated as isotropic: using the real head-space geometry roughly halves the prediction error relative to an isotropic null, so coordinate coupling matters.

A Chernoff bound makes the result structural rather than incidental: uniform componentwise accuracy at $\varepsilon = 0.1$ requires K of order 10^3–10^4, against an available budget of $K \leq r = 7$. The gap is two to three orders of magnitude, so no amount of tuning recovers the max-component rule from a small sketch.

6 Generality

6.1 Committee size

Enumerating *all* 247 subcommittees of every size across 20 systems, halving the committee to $m = 4$ retains 0.970 of Factor A on average. But the worst individual four-member subcommittee retains only 0.732, and at $m = 2$, 0.548. Committee size buys *insurance*, not accuracy: going from four heads to eight moves the mean by three points and the floor by twenty-seven.

6.2 Independently trained committees

Eight separately trained models, matched to shared-trunk counterparts on training data and evaluation frames, confirm the global-statistic recommendation—so it is a property of committees, not of the multi-head architecture. They are also far better calibrated in magnitude: overconfidence 2.28× against 25.77× for the matched shared-trunk committee. The *acceleration*, by contrast, does not transfer at all, and we state this as a derivation rather than a measurement: with no shared trunk, obtaining Fw requires all M columns of F, so no sketch can be cheaper than the exact quantity. Of five pre-registered predictions in this experiment, 3 were confirmed.

6.3 Condensed phase and distribution shift

Bulk liquid water ($N = 192$, periodic, $D = 576$) and a same-temperature MD-versus-PIMD pair give a distribution shift in which only quantum nuclear effects differ. One caveat must survive contact with the reader: the shift regime's striking Factor A is substantially a symptom of *this model family failing badly there.* A matched independently-trained committee is 2.7× more accurate on the shifted data and its Factor A collapses, while a free signal retains most of the apparent skill. Shift-regime claims must therefore be quoted beside the free-signal baseline and the accuracy gap.

7 Foundation models, real pools, and the baseline that must be beaten

The setting the method is actually pitched at is a foundation model with committee readouts on a frozen trunk [2]. We scan 136,923 periodic crystals drawn from MPtraj [5], spanning 89 elements and 1–444 atoms, with a MACE-MP [1] trunk fine-tuned to carry eight disjoint committee heads. Five predictions were pre-registered; two were confirmed.

7.1 Factor A transfers

Exact committee disagreement reaches AUROC 0.798 here against a random null of 0.499. Committee force uncertainty is not an artefact of small organic molecules.

7.2 Extensivity, and why the global statistic reverses

The global statistic sums v_d over $3N$ coordinates, so it grows with system size whether or not the model is uncertain. Acquiring by it returns structures whose median atom count is 2.50× that of the max-component selection, and the ordering of Section 3 reverses on all four error scores (Figure 4a, Table 1). Every system in Section 3 has fixed N, where an extensive and an intensive statistic differ

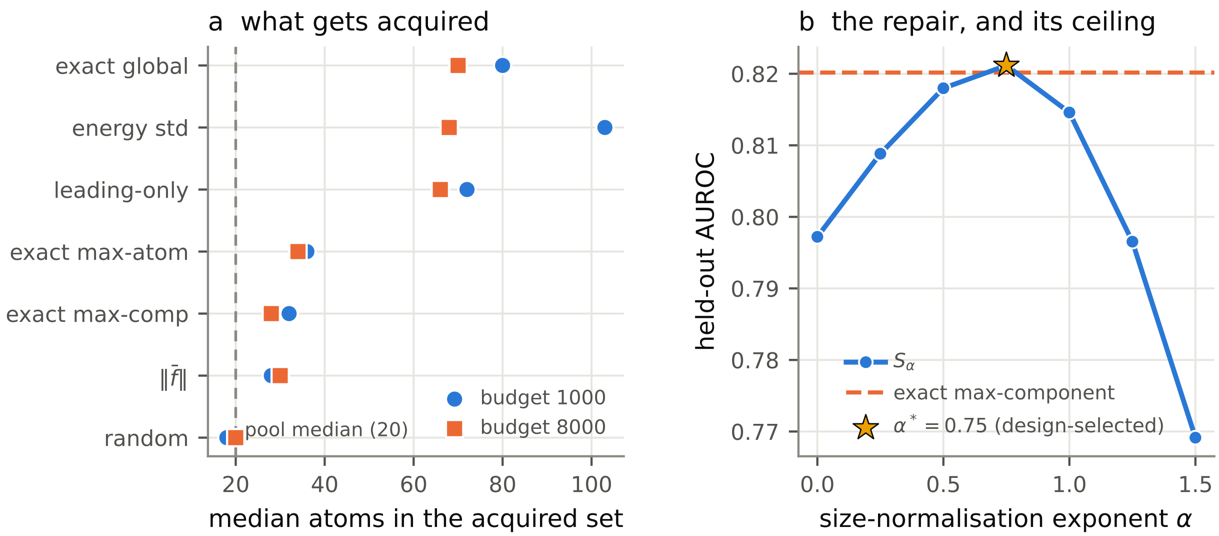

Figure 4: The global statistic is extensive, and normalisation restores parity rather than superiority. (a) Median atom count of the acquired set on the MPtraj pool. The global statistic and the two quantities that inherit its extensivity acquire far larger structures than the pool median; the intensive statistics and the free signal do not. (b) Held-out AUROC of $S_\alpha = (\sum_d v_d)/(3N)^\alpha$ against the exponent α, with the design-selected α^* starred. The curve peaks near max-component's level, not above it.

by one constant and the pathology *cannot* appear: the recommendation was never wrong where it was measured, it was measured only where it could not fail.

The repair is a size normalisation, $S_\alpha = (\sum_d v_d)/(3N)^\alpha$, with α chosen on a design split and every reported number taken from the disjoint remainder. The selected $\alpha = 0.75$ raises held-out AUROC from 0.797 to 0.821 (Figure 4b), and at fixed N it changes nothing exactly—so it can be adopted globally at no cost. We report the outcome precisely: across four error scores the normalised statistic achieves 2 significant wins, one tie, and 1 significant loss against max-component. This restores *parity*, not superiority, and α is pool- and target-dependent rather than universal.

7.3 The free signal wins here, and is not complemented

The max mean-force norm reaches AUROC 0.943 against the committee's 0.798—a signal that is already computed in the forward pass outranking eight backward passes (Table 1). The advantage survives stratification by atom count and, within force-magnitude quintiles, the committee still beats chance, so it does carry information beyond force magnitude; a free proxy is simply at least as good.

Because the free signal is free, the decision a practitioner faces is not "committee or free signal" but "free signal, or free signal *plus* eight backward passes". Nested models fitted on a design split and scored on held-out data give AUROC 0.9425 for the free signal alone and 0.9427 with the committee added—a marginal of 0.0002, not significant. On fixed-composition systems the same comparison gives significant gains on 8 of 10 systems, spanning 0.022–0.243. The committee is redundant *here*, not in general, and the discriminating variable is how close the free signal already is to saturation.

Table 1: On a real materials pool a free signal outranks every committee statistic. AUROC on 136, 923 MPtraj crystals under a frozen foundation-model trunk. The last two columns are post-hoc diagnostics: AUROC recomputed within atom-count quartiles and within force-magnitude quintiles, so neither the extensivity confound nor force magnitude itself can explain the ordering. $\|\bar{f}\|$ is bold because it is the signal that costs nothing.

	AUROC by error score				stratified (post hoc)	
signal	$e_{\max}$	e_{rmse}	$e_{\max\mathrm{c}}$	e_{q95}	by size	by force
exact max-atom	0.836	0.819	0.824	0.806	–	0.649
exact max-component	0.821	0.808	0.822	0.805	0.798	0.656
exact global	0.798	0.769	0.793	0.759	0.794	0.632
leading-only, 5 lanes	0.791	0.763	0.786	0.753	0.779	0.626
$\|\bar{f}\|$ *(free)*	**0.943**	**0.933**	**0.942**	**0.930**	**0.933**	**0.710**
energy std *(free)*	0.481	0.447	0.483	0.450	0.394	0.511
random null	0.499	0.498	0.498	0.498	–	0.499

Table 2: Pre-registered systems predictions, scored per device on matched pairs. Water and 3BPA measured in one exclusive job per device at identical settings. v2 is the frozen pre-registered protocol; v3 is a post-hoc repair that derives its reference from the matched 3BPA run rather than a fixed constant, and it changes the score only on Blackwell. P1 and P2 — the predictions carrying the speedup claim — pass everywhere.

	verdict		pre-registered predictions				speedup
device	v2	v3	P1	P2	P3	P4	water, B=16
H200	**4/4**	4/4	✓	✓	✓	✓	1.852×
A100-SXM4-80GB	**3/4**	3/4	✓	✓	×	✓	1.880×
RTX PRO 6000 Blackwell	**2/4**	3/4	✓	✓	×	×	1.894×

7.4 The selection-overlap bound is unavailable

One can hope to bound the downstream active-learning difference without retraining, by showing the gate acquires nearly the same set as the exact statistic. It does not: Jaccard overlap is 0.794 at a realistic acquisition rate, short of the pre-registered threshold. Overlap statistics must always be reported with their acquisition rate, since any two rankings agree when a large fraction of the pool is taken.

8 Wall clock

Lane reduction pays only once the backward pass is compute-bound rather than launch-bound, and the controlling variable is the total edge count in the batch, not the number of structures. Measuring both systems in a single exclusive job at identical settings, water reaches 1.53× at $B = 1$ rising to 1.88× at $B = 16$ on an A100, 1.85× on an H200 and 1.89× on an RTX PRO 6000 Blackwell—three GPU generations agreeing to 2.3%—against 1.10× for 3BPA in the same job (Figure 5, Table 2). Because the threshold is a property of the machine as well as the system, the batch size at which lane reduction begins to pay *increases* with how fast the GPU is.

Acceleration is deliberately not the headline. The count of avoided exact evaluations is hardware-independent and survives every timing outcome.

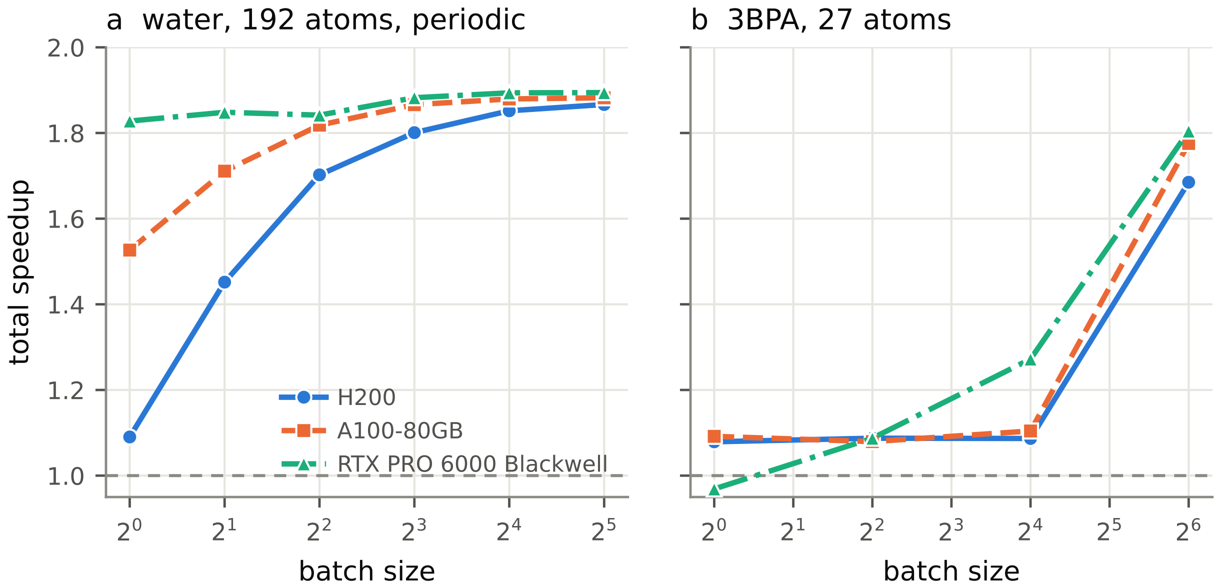

Figure 5: Lane reduction pays once the backward pass is compute-bound, and the batch size at which that happens rises with machine speed. Total speedup against batch size for (a) water and (b) 3BPA, measured with both systems in a single exclusive job per device. Water converges to 1.88×–1.89× by $B = 16$ on all three architectures; 3BPA stays near 1.10× except on Blackwell. The H200 is the fastest machine and is still launch-bound at $B = 1$.

9 Recommendations and limitations

1. **Score committee uncertainty against the free mean-force norm**, not against a random null. On some system classes it wins by a wide margin; on others it adds nothing measurable.

2. **Use the global statistic at fixed system size**; normalise by $(3N)^\alpha$ for variable size and refit α per pool and per target.

3. **Prefer a deterministic leading subspace to a randomised sketch**, at equal budget.

4. **Committee size buys insurance.** To fix magnitude calibration, change the construction, not M.

Limitations. One architecture family; $M = 8$ with subcommittee interpolation and no trained $M = 16$ control. No retraining was performed, so no claim is made about downstream model quality—only about which structures are selected. The selection-overlap bound is unavailable, so that gap cannot be closed cheaply. α is pool-dependent. Finally, several results here correct earlier statements of our own; the record of those corrections is retained in the repository rather than silently overwritten.

Data and Software Availability

All statistical results regenerate on CPU from repository-tracked artifacts: `tools/reproduce.py --tier offline` re-derives every headline quantity in roughly four minutes with no model execution, no CUDA and no download, and checks each against its committed value. `tools/audit.py` verifies the frozen prior-work tree, the sha256 binding of every pre-registered protocol to a registration record written before that experiment produced any artifact, the provenance of every number in this manuscript, the split manifests, and the test-access ledger. `tools/report_go_no_go.py` emits every scored verdict as data, alongside the confidence stated for it in advance.

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